Lab 3: Genetic Recombination in *Drosophila melanogaster*

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Abstract

In this experiment, we wanted to look at genetic recombination in *Drosophila melanogaster*. We used the GAL4/UAS transgene system to "rescue" two recessive traits. These traits were white eyes and yellow body. Before our experiment began, the GAs had already crossed flies carrying both recessive traits. One of the flies had the *UAS-RNAi*(y^+) transgene, whereas the other had the *UAS-Dcr-2(mw)* transgene. Our group crossed female flies from their progeny (F1) with wild-type males, and recorded the phenotypes of the F2 generation.

Introduction

Genetic recombination is the process by which genetic material is exchanged between two homologous chromosomes, by a process called crossing over. It is a fundamental process during meiosis that leads to genetic diversity. This exchange of genetic material results in the formation of new combinations of alleles, which are then passed on to the offspring.

In our experiment, we made use of the GAL4/UAS transgene system. This system is actually found in yeast. UAS stands for Upstream Activating Sequence, and GAL4 is a transcription factor that binds to UAS and activates gene expression. It is used for targeted gene expression, in a specific tissue, and under specific conditions.

In our setup, we tracked two separate transgenes: *UAS-RNAi*(y^+) and *UAS-Dcr-2(mw)*. The y^+ and *mw* (mini-white) sequences serve as selectable markers; in a *y w* mutant background, they rescue the brown body and orangish-red eye phenotypes, respectively, allowing us to track the inheritance of the transgenes.

The UAS-RNAi gene produces a double-stranded RNA (dsRNA) sequence complementary to a target gene's mRNA, leading to its degradation. To enhance this effect, we used UAS-Dcr-2, which expresses the Dicer-2 enzyme. Dicer-2 acts as molecular scissors, cleaving the dsRNA into small interfering RNAs (siRNAs), thereby making the gene silencing more potent and consistent.

To begin with, a fly with the *UAS-RNAi(y^+)* gene was crossed with a fly carrying the *UAS-Dcr-2(mw)* gene. The genes were on the autosomes, so they could have either been on the same chromosome, or on different chromosomes.

So, our job was to figure out the genotypes of the F0 flies, using recombination frequency. If the genes were on the same chromosome, but on different homologous chromosomes, they would recombine. If they were on different chromosomes all together, they would not recombine, and would assort independently instead.

Materials and Methods

Materials

The female flies we used from the F1 generation were of the following phenotype: orange eyes and brown body. The males were homozygous for the mutations that cause white eyes and yellow bodies, and they had stubble. They were kept in vials with cornstarch feed, and some yeast paste was applied to the surface of the feed. The flies were handled on a flypad, and anaesthetized using CO_2 gas. A fine brush was used to handle them, and they were observed under a dissection stereomicroscope.

Methods

In a cornstarch food vial, we applied a small amount of yeast paste to the surface of the feed to promote egg laying. We then added 6 female flies with the markers and placed them in the vial along with 3 males. We left the vials for 14 days to allow the flies to mate and lay eggs. On the 14th day, we collected the F2 progeny, anaesthetized them and recorded the number of flies and their phenotype.

Results

The phenotypic data collected from the F2 progeny is summarized in the table below. Table 1 shows the frequency of each trait combination.

Table 1: Combination Frequencies

EYE COLOUR	BODY COLOUR	COUNT
Orange	Yellow	46
White	Brown	42
White	Yellow	12
Orange	Brown	10
	Total	**110**

Discussion

Case 1: Transgenes are on different chromosomes

If the genes were on different chromosomes, they would assort independently.

The cross in the F0 generation would look like:

$$y\ w; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{+} \times y\ w; \frac{+}{+}; \frac{UAS-RNAi(y^+)}{+}$$

Meaning the F1 progeny would consist of:

$$y\ w; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{UAS-RNAi(y^+)}$$
$$y\ w; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{+}$$
$$y\ w; \frac{+}{+}; \frac{+}{UAS-RNAi(y^+)}$$
$$y\ w; \frac{+}{+}; \frac{+}{+}$$

So the phenotypes would be: orange eyes brown body, orange eyes yellow body, white eyes brown body, and white eyes yellow body, respectively.

We then chose the female flies from the F1 generation with the phenotype orange eyes and brown body for our next cross. We crossed these females with males that were recessive homozygous for white eyes and yellow body, and recorded the phenotypes of the F2 generation.

Assuming the transgenes were on different chromosomes, the cross would look like:

$$\frac{y\ w}{y\ w}; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{UAS-RNAi(y^+)} \times \frac{y\ w}{Y}; \frac{+}{+}; \frac{+}{+}$$

This would result in a 1:1:1:1 ratio of the following genotypes:

$$y\ w; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{UAS-RNAi(y^{+})}$$
$$y\ w; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{+}$$
$$y\ w; \frac{+}{+}; \frac{+}{UAS-RNAi(y^{+})}$$
$$y\ w; \frac{+}{+}; \frac{+}{+}$$

So the phenotypes would be: orange eyes brown body, orange eyes yellow body, white eyes brown body, and white eyes yellow body, respectively.

However, that is not what we observed.

Case 2: Transgenes are on the same chromosome

If the genes were on the same chromosome, but on different homologous chromosomes, the cross in the F0 generation would look like:

$$\frac{y\ w}{y\ w}; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{+} \times \frac{y\ w}{y\ w}; \frac{UAS-RNAi(y^{+})}{+}; \frac{+}{+}$$

Meaning the F1 progeny would consist of:

$$\frac{y\ w}{y\ w}; \frac{UAS-Dcr-2(mw)}{UAS-RNAi(y^{+})}; \frac{+}{+}$$
$$\frac{y\ w}{y\ w}; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{+}$$
$$\frac{y\ w}{y\ w}; \frac{+}{+}; \frac{UAS-RNAi(y^{+})}{+}$$
$$\frac{y\ w}{y\ w}; \frac{+}{+}; \frac{+}{+}$$

So the phenotypes would be: orange eyes brown body, orange eyes yellow body, white eyes brown body, and white eyes yellow body, respectively.

We then chose the female flies from the F1 generation with the phenotype orange eyes and brown body for our next cross. We crossed these females with males that were recessive homozygous for white eyes and yellow body, and recorded the phenotypes of the F2 generation.

Assuming the transgenes were on the same chromosome, the cross would look like:

$$\frac{y\ w}{y\ w}; \frac{UAS-Dcr-2(mw)}{UAS-RNAi(y^{+})}; \frac{+}{+} \times \frac{y\ w}{Y}; \frac{+}{+}; \frac{+}{+}$$

If no recombination took place, it would result in a 1:1 ratio of the following genotypes:

$$y\ w; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{+}$$
$$y\ w; \frac{UAS-RNAi(y^{+})}{+}; \frac{+}{+}$$

And, no parental phenotypes would be observed in the F2 generation. However, out of the 110 flies we collected, 20% of them had parental phenotypes. The recombination frequency was calculated as follows:

$$\text{Recombination Frequency} = \frac{\text{Number of Recombinant Offspring}}{\text{Total Number of Offspring}} \times 100$$

$$\text{Recombination Frequency} = \frac{22}{110} \times 100 = 20\%$$

This confirms that recombination has taken place and that the two transgenes are located on the same chromosome. So, the genotypes of the F0 generation were:

$$\frac{y\ w}{y\ w}; \frac{UAS-Dcr-2(mw)}{+}; \frac{+}{+} \times \frac{y\ w}{y\ w}; \frac{UAS-RNAi(y^{+})}{+}; \frac{+}{+}$$